Bonus Assignment

1. Given that,

$$W = \{(x_1, x_2, x_3, 0) : x_1, x_2 \text{ and } x_3 \text{ are real number}\}$$

$$V = \mathbb{R}^4$$

Let,

$$u = (u_1, u_2, u_3, 0) \in W$$

$$v = (v_1, v_2, v_3, 0) \in W$$

$$\therefore (u+v) = (u_1, u_2, u_3, 0) + (v_1, v_2, v_3, 0)$$

$$= (u_1 + v_1, u_2 + v_2, u_3 + v_3, 0) \in W$$

$$\therefore (cu) = c(u_1, u_2, u_3, 0)$$

$$= (cu_1, cu_2, cu_3, 0) \in W$$

So, W is a subspace of V. (Ans:)

2. Given that,

$$W = \{(x, y, 2x - 3y) : x \text{ and } y \text{ are real numbers}\}$$

$$V = \mathbb{R}^3$$

Let,

$$u = (u_1, u_2, 2u_1 - 3u_2) \in W$$

$$v = (v_1, v_2, 2v_1 - 3v_2) \in W$$

$$\therefore (u+v) = (u_1, u_2, 2u_1 - 3u_2) + (v_1, v_2, 2v_1 - 3v_2)$$

$$= (u_1 + v_1, u_2 + v_2, 2u_1 - 3u_2 + 2v_1 - 3v_2)$$

$$= \{u_1 + v_1, u_2 + v_2, 2u_1 + 2v_1 - (3u_2 + 3v_2)\} \in W$$

$$\therefore (cu) = c(u_1, u_2, 2u_1 - 3u_2)$$

$$= (cu_1, cu_2, 2cu_1 - 3cu_2) \in W$$

So, W is a subspace of V. (Ans)

3. Given that,

$$2\times 2 \text{ Matrix} = \begin{bmatrix} 0 & a \\ b & 0 \end{bmatrix}$$

$$V = M_{2,2}$$

Let,

$$u = \begin{bmatrix} 0 & u_1 \\ u_2 & 0 \end{bmatrix} \in W$$

$$v = \begin{bmatrix} 0 & v_1 \\ v_2 & 0 \end{bmatrix} \in W$$

$$\therefore (u+v) = \begin{bmatrix} 0 & u_1 \\ u_2 & 0 \end{bmatrix} + \begin{bmatrix} 0 & v_1 \\ v_2 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & u_1+v_1 \\ u_2+v_2 & 0 \end{bmatrix} \in W$$

$$\therefore (cu) = c \begin{bmatrix} 0 & u_1 \\ u_2 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & cu_1 \\ cu_2 & 0 \end{bmatrix} \in W$$

So, W is a subspace of V. Ans

4. Given that,

$$3\times 2 \text{ Matrix} = \begin{bmatrix} a & b \\ a+b & 0 \\ 0 & c \end{bmatrix}$$

$$V = M_{3,2}$$

let,

$$u = \begin{bmatrix} u_1 & u_2 \\ u_1+u_2 & 0 \\ 0 & u_3 \end{bmatrix} \in W$$

$$v = \begin{bmatrix} v_1 & v_2 \\ v_1+v_2 & 0 \\ 0 & v_3 \end{bmatrix} \in W$$

$$\therefore (u+v) = \begin{bmatrix} u_1 & u_2 \\ u_1+u_2 & 0 \\ 0 & u_3 \end{bmatrix} + \begin{bmatrix} v_1 & v_2 \\ v_1+v_2 & 0 \\ 0 & v_3 \end{bmatrix}$$

$\cancel{u_1+v_1 \qquad u_2+v_2}$

$$= \begin{bmatrix} u_1+v_1 & u_2+v_2 \\ u_1+u_2+v_1+v_2 & 0 \\ 0 & u_3+v_3 \end{bmatrix}$$

$$= \begin{bmatrix} u_1+v_1 & u_2+v_2 \\ (u_1+v_1)+(u_2+v_2) & 0 \\ 0 & u_3+v_3 \end{bmatrix} \in W$$

$$\therefore (cu) = \begin{bmatrix} cu_1 & cu_2 \\ cu_1+cu_2 & 0 \\ 0 & cu_3 \end{bmatrix} \in W$$

~~$\therefore$ W is a~~

So, W is a subspace of V. Ans.

Q-2: Let,

$$W = \{(0, 0, -1)\}$$

$$V = \mathbb{R}^3$$

$\therefore u = (0, 0, -1) \in W$

$\therefore v = (0, 0, -1) \in W$

Now,

$$(u+v) = (0+0, 0+0, -1-1)$$
$$= (0, 0, -2) \notin W$$

Again,

Let, $c = 10$

$$\therefore cu = 10 \times (0, 0, -1)$$
$$= (0, 0, -10) \notin W$$

$\therefore$ W is not a subset of V.

Ans.

Let,

$$W = \{(0,1)\}$$

$$V = \mathbb{R}^2$$

$\therefore u = (0,1) \in W$

$\therefore v = (0,1) \in W$

Now,

$$(u+v) = (0+0,\ 1+1)$$

$$= (0,2) \notin W$$

Again,

Let, $c = 10$

$\therefore cu = 10 \times (0,1)$

$= (0,10) \notin W$

$\therefore$ W is not a subset of V.

Ans.

Let,

$$W = \left\{ \left(\frac{1}{2}, \frac{1}{3} \right) \right\}$$

$$V = \mathbb{R}^2$$

$\therefore u = \left(\frac{2}{3}, \frac{5}{8} \right) \in W$

$\therefore v = \left(\frac{1}{3}, \frac{1}{8} \right) \in W$

Now,

$$(u+v) = \left(\frac{2}{3} + \frac{1}{3}, \frac{5}{8} + \frac{1}{8} \right)$$

$$= \left(1, \frac{6}{8} \right) \notin W$$

Again, Let, $c = 2$

$$\therefore cu = 2 \times \left(\frac{2}{3}, \frac{5}{8} \right)$$

$$= \left(\frac{4}{3}, \frac{10}{8} \right) \notin W$$

$\therefore$ W is not a subset of V.

Let,

$$W = \{(1, 1)\}$$

$$V = \mathbb{R}^2$$

$\therefore u = (2, 2) \in W$

$\therefore v = (3, 3) \in W$

Now,

$$(u+v) = (2+3, 2+3)$$

$$= (5, 5) \notin W$$

Again,

Let, $c = 5$

$$\therefore (cu) = 5 \times (2, 2)$$

$$= (10, 10) \notin W$$

So, W is not a subset of V.

Ans.